Solution

(i) $(-35) \times (-11) = 385$ (ii) $96 \times (-20) = -1920$

(iii) $(-5) \times 12 = -60$ (iv) $15 \times 5 = 75$

(v) $999 \times 0 = 0$

Example 1.18 A fruit seller sold 5kg of mangoes at a profit of ₹ 15 per kg and 3kg of apples at a loss of ₹ 30 per kg. Find whether it is a profit or loss.

Solution

Profit of 1kg mangoes = ₹ 15

Profit of 5kg mangoes = 15 × 5

= ₹ 75

Loss of 1kg apples = ₹ 30

Loss of 3kg apples = 30 × 3

= ₹ 90

Loss = 90 – 75

= ₹ 15

Example 1.19 Browsing rates in an internet centre is ₹ 15 per hour. Nila works on the internet for 2 hours in a day for 5 days in a week. How much does she pay?

Solution

Number of hours spent on an internet for a day = 2 hrs

Therefore, number of hours spent on the internet for 5 days = 5 × 2

= 10 hrs

Cost of browsing per hour = ₹ 15

Cost of browsing for 10 hours = 15 × 10

Therefore, the amount paid by Nila for 5 days = ₹ 150

1.4.1 Properties of Multiplication of Integers

Recall that multiplication of whole numbers had the closure property. If we test this property for integers we find $(-7)\times(-2) = +14$, $(-6)\times 5 = -30$, $4\times(-9) = -36$. Thus, product of any two integers is always an integer. "Closure property" holds for integers.

FHKLQ

Therefore, for any two integers a, b ; $a\times b$ is also an integer.

Consider the examples, $21\times(-5) = -105$ is the same as $(-5)\times 21 = -105$. The product $(-9)\times(-8) = +72$, $(-8)\times(-9) = 72$. In both the cases the product is the same integer. So, changing the order of multiplication does not change the value of the product. Hence, we conclude that multiplication of integers are "commutative".

Therefore, for any two integers a, b ; $a\times b = b\times a$.

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